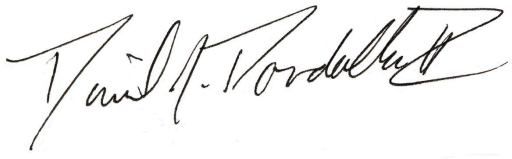
The role of a novel artificial reef and marine protected areas in the recovery of threatened echinoderm populations along the Palos Verdes Peninsula

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Abstract

An outbreak of wasting disease, triggered by a warm water anomaly known as 'The Blob', caused a series of mass mortality events in sea star, urchin, and cucumber populations along the Southern California Bight throughout 2013-2016. Each taxon suffered from various forms of wasting syndrome (WS), initially attributed to a viral infection that worsens during periods of elevated water temperature. Heightened immune stress increased susceptibility to pathogenic infection, negatively impacting recovery efforts. Two potential sources of refugia, artificial reef structures, and marine protected areas, potentially exhibit greater resilience and allow for echinoderm recovery due to higher nutrient availability, cool-water upwelling, improved habitat quality, and protection from fishing. By analyzing time series trends and employing probable variability models assessing the response to environmental impacts (progressive-change BACIPS), the study evaluates the recovery status of ochre sea stars (*Pisaster ochraceus*), bat stars (*Patiria miniata*), giant-spined sea stars (*P. giganteus*), purple sea urchins (*Stronglyocentrotus purpuratus*), red sea urchins (*Mesocentrotus franciscanus)*, warty sea cucumber (*Apostichopus parvimensis*), and the giant California sea cucumber (*A. californicus*). Natural reef sites adjacent to newly constructed artificial reef structures are preferable habitats for affected taxa compared to those within marine protected areas. These findings uncover the effectiveness of conservation strategies in mitigating the impacts of environmental disturbances on vulnerable echinoderm populations.

Introduction

Sea stars, urchins, and cucumber (Echinodermata) populations collapsed in the past decade due to an outbreak of wasting syndrome/disease (WS) in kelp forest ecosystems from British Columbia to Southern California [1,2]. Each taxon suffered from varied forms of WS, the origins of which were initially thought to be brought on by viral infection. WS has now come to represent heterogeneous etiologies, most notably elevated water temperature and variation in environmental conditions[3,4]. During extreme climate periods, echinoderm populations suffered from heightened susceptibility to pathogens resulting from immune stress, and worsening signs of wasting[5]. A period of warm water anomalies, known as 'The Blob', lasted from 2013-16 and coincided with a particularly strong El Niño event that resulted in a depletion of nutrients and high sea surface temperatures along the Southern California Bight[5]. These anomalies are occurring more frequently as the impacts of climate change are realized[5]. The effects of the Blob resulted in extensive ecological damage to rocky reef ecosystems characterized by frequent mortality events of seagrass and kelp[6]. During this period, multiple mass mortality events of echinoderm populations were recorded, especially along the Palos Verdes Peninsula[6,7].

These taxa, with stars and urchins being known as 'keystone predators', exert a substantial influence on driving biodiversity, population dynamics and ecological shifts within rocky reef communities[8]. WS moderates the strength of the affected populations' ecological interactions, disrupting the hosts' prey consumption, increasing predation vulnerability, and even directly causing death[3,4]. The most prevalent WS is wasting asteroid-associated densovirus (WAaDs), which causes loss of turgor, bodily lesions, and dissolution of limbs for sea stars[2]. Similarly, sea urchins suffer from wasting/black spot disease, which causes atypical morphological signs, such as loss of spines, white lesions, and black spots on the test[9]. Sea

cucumbers, rather, suffer from skin ulceration and evisceration[10]. Since its first discovery in 1978, sea star and urchin populations have suffered many mass mortality events due to WS, with the most recent during 2013-14[4]. In sites throughout Oregon, sea star densities declined by 50-94%, and in the San Juan Islands, up to 25% of their pre-wasting abundance[4]. Of the taxa in this study, *P. ochraceous,* the ochre sea star, exhibited the most severe population decline since 2014-15[11]. Overall, the impact of WS on echinoderms significantly disrupted rocky reef communities, highlighting the need for conservation efforts to mitigate the effects on biodiversity and ecology stability.

Organisms susceptible to environmental disturbances elicit better recovery when they inhabit or are near areas with high-quality habitats, such as artificial reefs and marine protected areas[12]. Marine-protected areas are conservation strategies that preserve biodiversity and at-risk populations, providing long-term protection against climate impacts[12]. Upcoast from PVR is the Point Vicente State Marine Conservation Area (SMCA), and the Abalone Cove SCMA. In these 'no take' marine protected areas the injuring, damaging, taking or possession of any echinoderm species from either conservation area is strictly prohibited[17]. Similarly to habitats near artificial reefs, this expansive MPA allows for the inclusion of cool water habitats, enhancing its ecological diversity, nutrient upwelling, and effectiveness, with the additional legal protection of marine species removing the anthropogenic pressure of industrial fishing. The habitats associated with this MPA consist of kelp forests, rocky reefs and hard and soft seafloor habitats, similar to that seen at the natural reef. Both MPAs were established in January 2012, one year before the Blob was observed. In 2020, the Vantuna Research Group (VRG) constructed the Palos Verdes Reef (PVR), a quarry rock reef on the Palos Verdes Peninsula to restore habitat that was buried by landslides. Such landslides have covered the below rocky reef habitat in chronic

sedimentation, depleting or displacing most invertebrate life and available nutrients provided by nitrogen, carbon, and phosphorus cycling from photosynthetic organisms in the ecosystem[13–15]. It is located adjacent to the existing natural rocky reef and kelp forest near the edge of the continental shelf. The proximity to the shelf allows for increased exposure to cool, nutrient-rich water upwelling from internal tides, making this location ideal for echinoderm recruitment and potential refugia for existing populations[14]. During initial planning for the reef, stacked species distribution modelling (sSDM) was utilized to identify the most ideal location for habitat restoration along the Southern California Bight[16]. The high-relief modules that make up PVR act as essential habitats for fish and substrate for giant kelp, other marine algae, and marine invertebrates. In a preliminary study, there was a positive response in fish density, fish biomass, kelp density, and biotic benthic cover at PVR post-construction, indicating its potential for enhanced recovery and resilience of marine ecosystems against environmental disturbance[14].

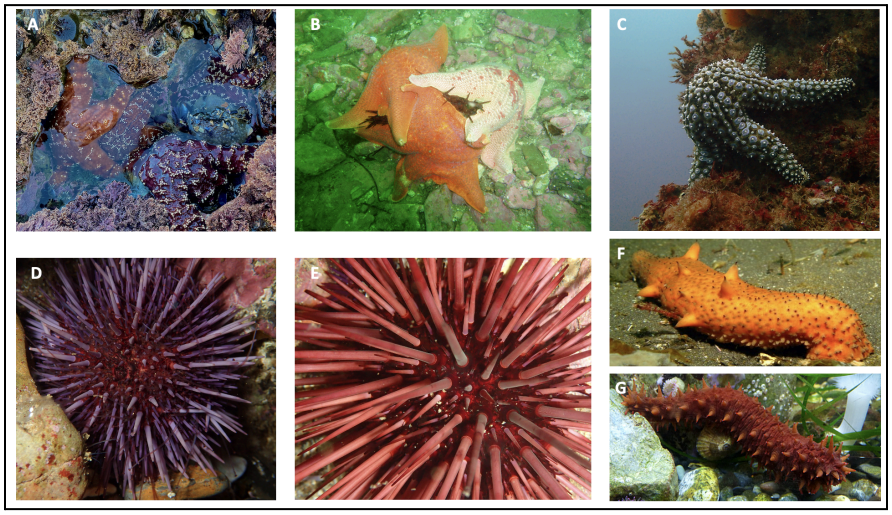


Figure 1. Images of focal species. (A) *P. ochraceous,* (B) *P. miniata,* (C) *P. giganteus,* (D) *S. purpuratus,* (E) *M. franciscanus,* (F) *A. parvimensis,* (G) *A. californicus*. Photos taken by A. Player (A), J. Williams (B-E, G), Richard Ling/flicker (F).

I examined the mean population density of ochre sea stars (*Pisaster ochraceus*), bat stars (*Patiria miniata*), giant-spined sea stars (*P. giganteus*), purple sea urchins (*Stronglyocentrotus purpuratus*), red sea urchins (*Mesocentrotus franciscanus*), warty sea cucumber (*Apostichopus parvimensis),* and the giant California sea cucumber (*A. californicus)* to analyse the role of natural reef structures adjacent to artificial reef habitat, PVR, and MPA habitats in the long-term recovery of these taxa in response to the environmental disturbance warming event, 'The Blob'. (Fig. 1). These species are susceptible to mass mortality and disease outbreaks in habitats experiencing temperature fluctuation. In addition, the red urchin is one of the top four most valuable invertebrate fishery species in southern California, with the warty sea cucumber ranking as one of the top two emergent invertebrate fishery species[3]. I tested whether or not the implementation of artificial reef habitats is an effective rehabilitation effort, as well as being a long-term refugia for these threatened echinoderm populations. By using the MPA as a proposed long-term model of a resilient habitat, the following predictions can be made:

PALOS VERDES REEF-ADJACENT REEFS AND NATURAL REEF

H_0: There is no significant difference in the mean species density of echinoderm populations in response to a wasting disturbance in natural reef habitats adjacent to artificial reefs compared to non-adjacent reef habitats.

H_A: There is a significant difference in the mean species density of echinoderm populations in response to a wasting disturbance in natural reef habitats adjacent to artificial reefs compared to non-adjacent reef habitats.

MARINE-PROTECTED AREA AND NATURAL REEF

H_0: There is no significant difference in the mean species density of echinoderm populations at the MPA compared to the natural reef following the wasting disturbance.

H_A: There is a significant increase in the mean species density of echinoderm populations at the MPA compared to the natural reef following the wasting disturbance.

Methods

Scuba divers visually surveyed each site annually from 2008 to 2023 at four depth zones (~5m, ~10m, ~15m, and ~20m) using modified CRANE (Cooperative Research and Assessment of Nearshore Ecosystems) protocols[13]. The rocky reef ecosystem characteristics in this analysis were characterized using 30 m × 2 m band transects (swath) to estimate the density of macroinvertebrates and kelp.

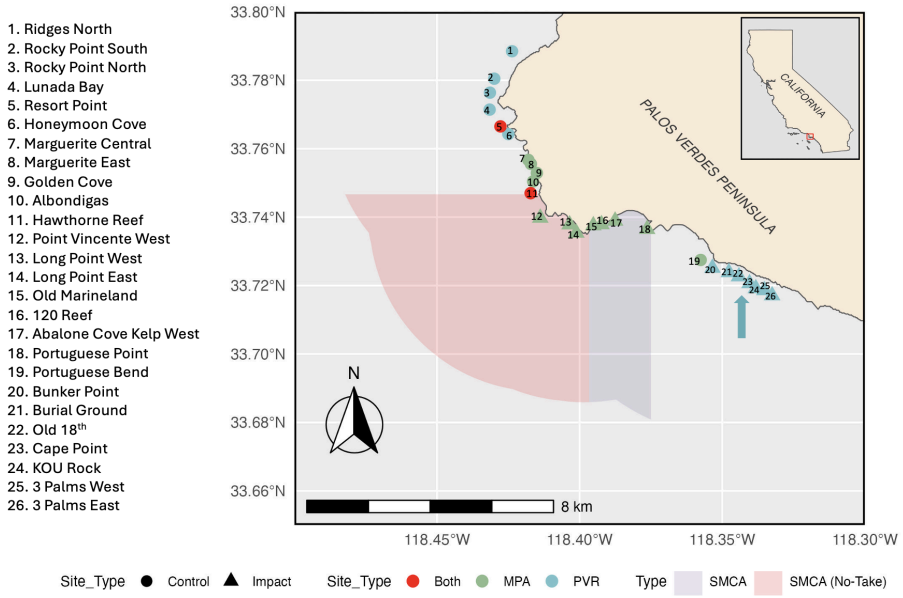


Figure 2. Map of sites surveyed on SCUBA used in both analyses. Sites are located on Palos Verdes Peninsula, Southern California as indicated by the locator map in the top right corner. Control sites are indicated by circles and impact sites are indicated by triangles. Colours indicate category type (green = MPA, blue = PVR-Adj, red = used in both analyses). Pink and purple areas show the State Marine Conservation Areas (SCMA 'MPAs'). PVR artificial reef modules are indicated by a blue arrow.

In the context of this paper, both conservation areas were considered one marine protected area (MPA) as the habitats are contiguous and have identical restrictions for the species included in this analysis. Paired sites were determined using the best professional judgement based on the similarity of habitat structure, oceanographic perimeters and physical characteristics (Table 1). Sites for the density analyses are the paired sites for the progressive-change BACIPS (before-after control-impact paired series) analyses (n=26) (Fig. 2). Annual mean density (per 100 m^2) of echinoderm focal species was analysed using density trends at 1) PVR-adjacent (PVR-Adj) and non-adjacent natural reef (NR) sites, and 2) MPA NR and non-MPA NR sites. Progressive-change BACIPS distinguish natural spatial and temporal variability from variation induced by an environmental impact[18]. The impact of this analysis for all sites is the wasting event of 2013-16, allowing for 5 years pre- and 7 years post-disturbance. Control sites for all analyses were selected from natural reef habitats unaffected by man-made implementation (PVR) or restoration efforts (MPA). Impact sites for PVR-Adj analysis have been chosen from sites adjacent to the artificial reef modules. Impact sites for the MPA-natural reef analysis are from survey sites within the marine protected area.

Table 1. Sites analysed in the progressive-change BACIPS analysis. Rows are aligned to show site parings.

PVR Analysis		MPA Analysis	
Control (Natural Reef)	**Impact (PVR-Adj)**	**Control (Non-MPA Natural Reef)**	**Impact (MPA Natural Reef)**
Hawthorne Reef	KOU Rock	Hawthorne Reef	Long Point East
Honeymoon Cove	Old 18th	Marguerite Central	120 Reef
Lunada Bay	Burial Grounds	Albondigas	Abalone Cove Kelp West
Resort Point	Cape Point	Marguerite East	Long Point West
Rocky Point South	3 Palms West	Golden Cove	Old Marineland
Rocky Point North	3 Palms East	Resort Point	Point Vicente West
Ridges North	Bunker Point	Portuguese Bend	Portuguese Point

Progressive-change BACIPS fit competing models to the spatiotemporal response of a system following an intervention. The output of the models included two of four possible dynamical responses seen in this analysis. In the linear model, differences change over time at a constant rate, whereas a step-change model portrays an instantaneous difference that remains constant over time[18]. An increase in the impact event is indicated by a positive delta value, whereas an increase in the control event is indicated by a negative delta value[18]. The statistical probability of the chosen model compared to another was determined by relative likelihood percentages (S1). Data from band (swath) transects was used to generate the density analyses and to perform the progressive-change BACIPS in R (Version 2023.12.0+369).

RESULTS

PALOS VERDES REEF

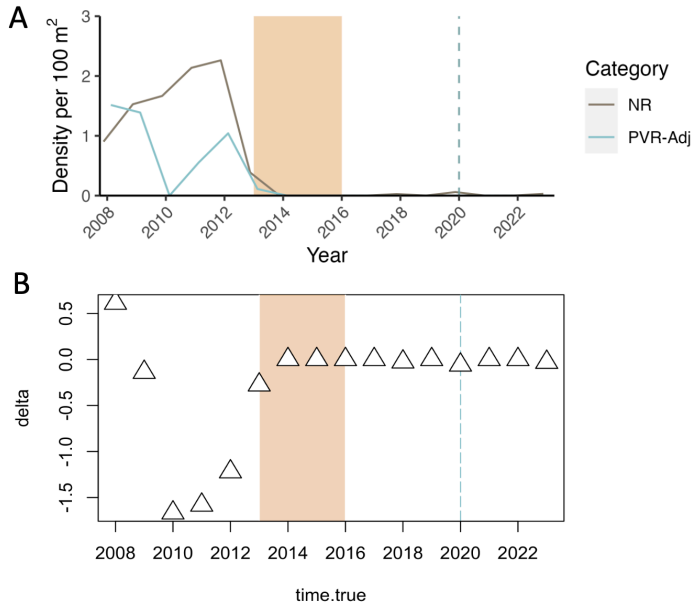


Figure 3. (A) Annual mean density per 100 m^2 of *P. ochraceus* from 2008-2023 at the natural reef (NR) and Palos Verdes Reef adjacent natural reef (PVR-Adj) sites before and after artificial reef module implementation in 2020. (B) Best-fit spatiotemporal model for *P. ochraceus* based on progressive-change BACIPS method. Blue dashed lines indicate PVR artificial reef module implementation in 2020. Orange bars indicate the Blob warming period and wasting event from 2013-2016.

Annual mean density for *P. ochraceus* at NR sites increased from 0.9/100 m^2 to 2.5/100 m^2 from the years 2008 through 2012. A sharp decline occurred from 2013 to 2014 when annual mean density dropped to 0.0/100 m^2, persisting through the Blob with no signs of recovery observed through 2023. At PVR-adjacent sites, the annual mean density was 1.5/100 m^2 in 2008, dropping to 0/100 m^2 in 2010. There was an increase to 1/100 m^2 from 2011 to 2012, remaining below 0.06/100 m^2 in 2013 through 2023. (Fig. 3A). Following the wasting event, the annual mean density of *P. ochraceous* declined abruptly (Fig. 3B), resulting in the selection of the step-change model (ω_{step} = 80.5%, *P*-value = 0.020). *P. ochraceus* did not recover at either habitat following the Blob, indicating that each habitat is acting the same in response to a disturbance.

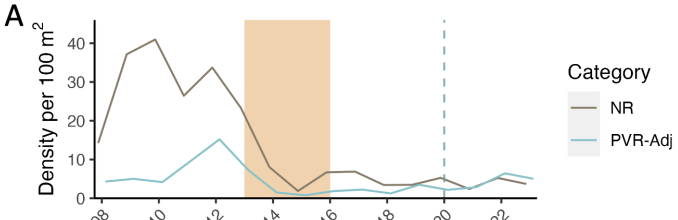


Figure 4. (A) Annual mean density per 100 m^2 of *P. miniata* from 2008-2023 at the natural reef (NR) and Palos Verdes Reef adjacent natural reef (PVR-Adj) sites before and after artificial reef module implementation in 2020. (B) Best-fit spatiotemporal model for *P. miniata* based on progressive-change BACIPS method. Blue dashed lines indicate PVR artificial reef module implementation in 2020. Orange bars indicate the Blob warming period and wasting event from 2013-2016.

Annual mean density fluctuated between 14.2/100 m^2 to 40.9/100 m^2 from 2008-2012. In 2013-2016, mean density declined to 6.7/100 m^2. *P. miniata* exhibited a minor increase in mean density in the following years and remained low, not exceeding 6.8/100 m^2. At PVR-adjacent sites, the annual mean density was between 4.1-4.3/100 m^2 from 2008 to 2010 before peaking at 15.2/100 m^2 in 2012. Annual mean density then declined below 7.3/100 m^2 from the years 2013 to 2020 when reef modules were implemented and then started to show an upward trend, surpassing the mean density of *P. miniata* in 2022 (Fig. 4A). Following the wasting event, the annual mean density of *P. miniata* declined abruptly (Fig. 4B), resulting in the selection of the step-change model (ω_{step} = 61.7%, *P*-value = 0.000). There was no significant difference in response between habitats for *P. miniata* in response to an environmental disturbance. Notably, *P. miniata* was a more resilient species and has made a minor recovery since the highest decline in density in 2015.

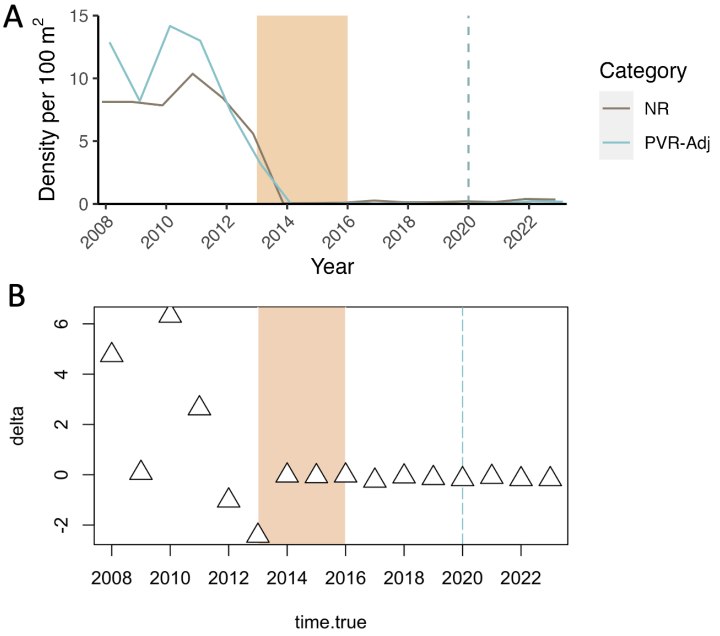


Figure 5. (A) Annual mean density per 100 m^2 of *P. giganteus* from 2008-2023 at the natural reef (NR) and Palos Verdes Reef adjacent natural reef (PVR-Adj) sites before and after artificial reef module implementation in 2020. (B) Best-fit spatiotemporal model for *P. giganteus* based on progressive-change BACIPS method. Blue dashed lines indicate PVR artificial reef module implementation in 2020. Orange bars indicate the Blob warming period and wasting event from 2013-2016.

The annual mean density of *P. giganteus* fluctuated at both PVR-adjacent and NR sites from 2008 to 2013, with minor annual fluctuation until 2013 where mean density sharply declined to 0.5/100 m^2 from 2011 through 2023 (Fig. 5A). Following the wasting event, *P. giganteus* were absent (Fig. 5A, B), resulting in the selection of the step-change model (ω_{step} = 94.3%, *P*-value = 0.009). The step-change model is statistically significant and can be accepted as the likely progression model in upcoming years. There was no significant difference in response to disturbance between these sites.

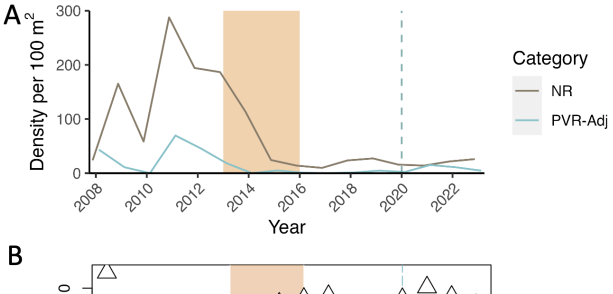


Figure 6. (A) Annual mean density per 100 m^2 of *S. purpuratus* from 2008-2023 at the natural reef (NR) and Palos Verdes Reef adjacent natural reef (PVR-Adj) sites before and after artificial reef module implementation in 2020. (B) Best-fit spatiotemporal model for *S. purpuratus* based on progressive-change BACIPS method. Blue dashed lines indicate PVR artificial reef module implementation in 2020. Orange bars indicate the Blob warming period and wasting event from 2013-2016.

The annual mean density of *S. purpuratus* at the NR sites varied between 23.8/100 m^2 and 287.6/100 m^2 prior to 2013. During the onset of the Blob, mean density decreased to 9.8/100 m^2 remaining through 2023. At PVR-adjacent sites, the mean density of *S. purpuratus* declined from 43.3/100 m^2 to 0.0/100 m^2 from 2008-2010. Another decline followed from 69.5/100 m^2 to 0.0/100 m^2 from 2011-2014. Annual mean density remained at or near zero until 2020 when the artificial reef modules are implemented at PVR. Mean density was 15.5/100 m^2 in 2021 before declining again towards 4.6/100 m^2 in subsequent years (Fig. 6A). Post-wasting, the annual mean density of *S. purpuratus* declined abruptly (Fig. 6B), resulting in the selection of the linear-change model (ω_{linear} = 75.1%, *P*-value = 0.000). The linear-change model is statistically significant and suggests that *S. purpuratus* populations will plateau at low densities in all-natural reef habitats. PVR-adjacent natural reef sites and NR sites did not respond significantly to disturbance.

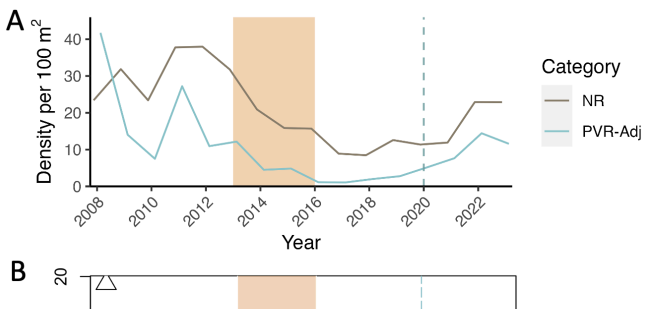


Figure 7. (A) Annual mean density per 100 m^2 of *M. franciscanus* from 2008-2023 at the natural reef (NR) and Palos Verdes Reef adjacent natural reef (PVR-Adj) sites before and after artificial reef module implementation in 2020. (B) Best-fit spatiotemporal model for *M. franciscanus* based on progressive-change BACIPS method. Blue dashed lines indicate PVR artificial reef module implementation in 2020. Orange bars indicate the Blob warming period and wasting event from 2013-2016.

At NR sites, the annual mean density of *M. francisanus* varied between 23.4/100 m^2 and 37.8/100 m^2 prior to 2013 before steadily declining to 8.8/100 m^2 in 2017. An upward trend of *M. francisanus* density was observed at NR sites with annual mean density increasing to 22.8/100 m^2 in 2023 (Fig. 7A). Following the wasting event, the annual mean density of *M. franciscanus* declined gradually (Fig. 7B), resulting in the selection of the linear-change model (ω_{linear} = 56.6%, *P*-value = 0.479). This model was not statistically significant and the next best model was step-change (ω_{step} = 42.0%). Both site categories exhibited a similar response post-disturbance.

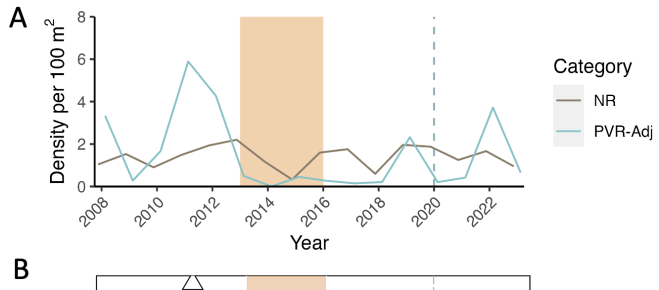


Figure 8. (A) Annual mean density per 100 m^2 of *A. parvimensis* from 2008 – 2023 at the natural reef (NR) and Palos Verdes Reef (PVR) sites before and after artificial reef module implementation in 2020. (B) Progressive-change BACIPS model for *A. parvimensis* showing probable spatiotemporal model. (C) Density overlay on PVR modules. Blue dashed lines indicate PVR implementation in 2020. Orange bars indicate the Blob warming period and wasting event from 2013/16.

The annual mean density of *A. parvimensis* at NR sites had no overall significant trend over the 16-year study period as mean density fluctuated between 1.0 and 0.9/100 m^2 from 2008 through 2023. For the years during the wasting disturbance, density decreased from 2.2/100 m^2 to 0.3/100 m^2 from 2013 to 2015, indicating a negative effect on cucumber density. There was a variation in mean density in 2016, fluctuating around 2.0/100m^2. At PVR-adjacent sites, the annual mean density was highest in 2011 (5.8/100 m^2). From 2012 through 2014, the mean density declined to 0.0/100 m^2. Following the implementation of artificial reef modules at PVR in 2020, the annual mean density of *A. parvimensis* increased to 3.7/100 m^2 from 2021 to 2022 before declining in 2023 to 0.6/100 m^2 (Fig. 8A). Post-wasting, the annual mean density of *A. parvimensis* had a slight decline (Fig. 8B), resulting in the selection of the step-change model (ω_{step} = 95.7%, *P*-value = 0.012). The natural reef and PVR-adjacent sites are showing no clear responses following the wasting event, and rather are exhibiting annual variation.

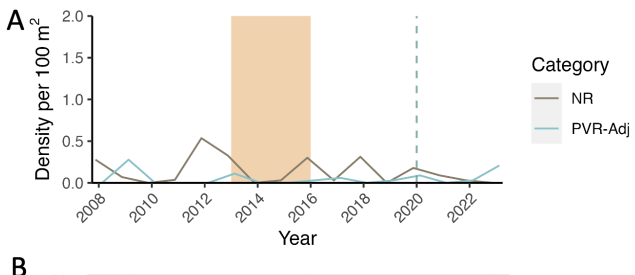


Figure 9. (A) Annual mean density per 100 m^2 of *A. californicus* from 2008-2023 at the natural reef (NR) and Palos Verdes Reef adjacent natural reef (PVR-Adj) sites before and after artificial reef module implementation in 2020. (B) Best-fit spatiotemporal model for *A. californicus* based on progressive-change BACIPS method. Blue dashed lines indicate PVR artificial reef module implementation in 2020. Orange bars indicate the Blob warming period and wasting event from 2013-2016.

The annual mean density of *A. californicus* was below 0.75/100 m^2 throughout the study period. At NR sites, mean density fluctuates annually, with the highest value being recorded at 0.52/100 m^2 in 2013 before declining to 0/100 m^2 in 2014 and 2015. A similar effect was observed at PVR-adjacent sites, with no *A. californicus* seen in 2011. However, following the implementation of modules at PVR in 2020, annual mean density increased from 0 to 0.25/100 m^2 from 2021 to 2023 which had not been observed at that level since 2008 (Fig. 9A). Following the wasting events, the annual mean density of *A. californicus* exhibited a slight decline (Fig. 9B), resulting in the selection of the linear-change model (ω_{linear} = 63.3%, *P*-value = 0.244). The linear-change model is not statistically significant, the next best model is step-change (ω_{step} = 33.9%). Both site categories are acting similarly, thereby supporting the null hypothesis.



Figure 10. (A) Annual mean density per 100 m^2 of *P. ochraceus* from 2008–2023 at the non-MPA natural reef (Non-MPA NR) and Marine Protected Area natural reef (MPA NR) sites before and after the wasting event of 2013-2016. (B) Best-fit spatiotemporal model for *A. californicus* based on progressive-change BACIPS method. Green dashed lines indicate MPA imposition in 2012. Orange bars indicate the Blob warming period and wasting event from 2013-16.

At non-MPA NR sites, the annual mean density of *P. ochraceous* varied between 2.5/100 m^2 and 10.2/100 m^2 prior to 2014 when it declined to 0.0/100 m^2 and did not recover through 2023. For MPA NR sites, the annual mean density of *P. ochraceous* increased from 15.7 to 17.7/100 m^2 from 2008 to 2009. Mean density declined in the following year dropping to 7.1/100 m^2 in 2010. With some annual fluctuation, the annual mean density declined through 2014 where it plateaued at 0.0/100 m^2 until 2023 (Fig. 10A). Following the wasting event, the annual mean density of *P. ochraceous* declined abruptly (Fig. 10B), resulting in the selection of the step-change model (ω_{step} = 66.4%, *P*-value = 0.028). Neither site category had an increase in annual mean density thereby supporting the null hypothesis for the MPA NR.

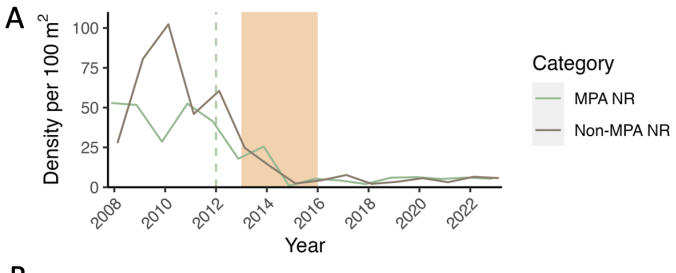


Figure 11. (A) Annual mean density per 100 m^2 of *P. miniata* from 2008–2023 at the non-MPA natural reef (Non-MPA NR) and Marine Protected Area natural reef (MPA NR) sites before and after the wasting event of 2013-2016. (B) Best-fit spatiotemporal model for *P. miniata* based on progressive-change BACIPS method. Green dashed lines indicate MPA imposition in 2012. Orange bars indicate the Blob warming period and wasting event from 2013-16.

The annual mean density of *P. miniata* at non-MPA sites increased from 27.7/100 m^2 to 102.3/100 m^2 from 2008-2010 before declining to 2.3/100 m^2 in 2015. Following this, annual mean density does not increase above 7.6/100 m^2 from 2016-2023 at non-MPA NR sites. At MPA NR sites, annual mean density varied between 28.6/100 m^2 and 52.9/100 m^2 before 2014 before declining to 5.3/100 m^2 in 2016. The annual mean density at MPA NR sites minorly increases from 2015-2023 but remains below 6.3/100 m^2 (Fig. 11A). Following the wasting event, the annual mean density of *P. miniata* declined gradually (Fig. 11A, B), resulting in the selection of the step-change model (ω_{step} = 67.2%, *P*-value = 0.116). The next best model was the asymptotic-change model ($\omega_{asymptotic}$= 62.8%), inferring a monotonous rate of change to 0. Both site categories acted similarly and annual mean density shows no significant recovery in response to wasting.

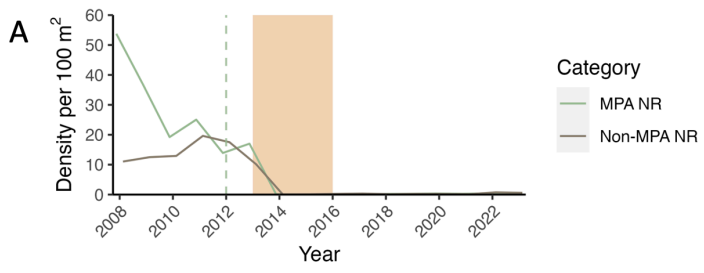


Figure 12. (A) Annual mean density per 100 m^2 of *P. giganteus* from 2008–2023 at the non-MPA natural reef (Non-MPA NR) and Marine Protected Area natural reef (MPA NR) sites before and after the wasting event of 2013-2016. (B) Best-fit spatiotemporal model for *P. giganteus* based on progressive-change BACIPS method. Green dashed lines indicate MPA imposition in 2012. Orange bars indicate the Blob warming period and wasting event from 2013-16.

At non-MPA NR sites, annual mean density varied between 11.0/100 m^2 and 19.2/100 m^2 from 2009-2013 before declining to 0.0/100 m^2 from 2014-2022. Density increases in 2023 to 0.6/100 m^2. For MPA NR sites, there was an initial high observation of *P. giganteus* at 53.8/100 m^2 in 2008. There was an overall decline in annual mean density from 2008 to 2014, at which point density plateaued below 1.0/100 m^2 through 2023 (Fig. 12A). Following the wasting event, the annual mean density of *P. giganteus* declined suddenly (Fig. 12B), resulting in the selection of the step-change model (ω_{step} = 77.9%, *P*-value = 0.013). This model was statistically significant and infers no recovery in upcoming years. There was no increase in annual mean density at either the MPA NR or NR sites, indicating that the MPA NR did not have a refugia effect in response to wasting.

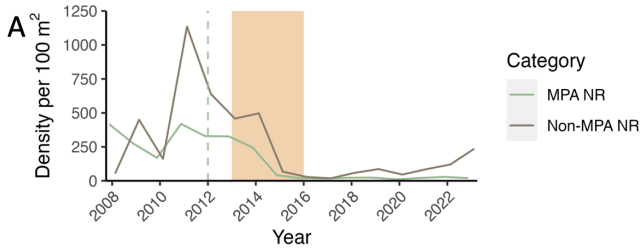


Figure 13. (A) Annual mean density per 100 m^2 of *S. purpuratus* from 2008–2023 at the non-MPA natural reef (Non-MPA NR) and Marine Protected Area natural reef (MPA NR) sites before and after the wasting event of 2013-2016. (B) Best-fit spatiotemporal model for *S. purpuratus* based on progressive-change BACIPS method. Green dashed lines indicate MPA imposition in 2012. Orange bars indicate the Blob warming period and wasting event from 2013-16.

The annual mean density for *S. purpuratus* at non-MPA NR sites was 53.3/100 m^2 in 2008, increasing over the following years to a peak of 1134.5/100 m^2 in 2011. At this point, the annual mean density declined to 17.9/100 m^2 in 2017, though it increased annually to 236.2/100 m^2 from 2017-2023. At MPA NR sites, annual mean density varied between 275.4/100 m^2 and 418.9/100 m^2 prior to 2013. From 2014-2015, annual mean density declined from 244.4/100 m^2 to 40.9/100 m^2, remaining below 86.2/100 m^2 from 2015 through 2023 with a minor upward trend (Fig. 13A). Following the wasting event, the annual mean density of *S. purpuratus* declined abruptly (Fig. 13B), resulting in the selection of the step-change model (ω_{step} = 52.4%, *P*-value = 0.511). The next best model was linear-change (ω_{linear} = 46.3%). The annual mean density of *S. purpuratus* showed signs of recovery at the non-MPA NR. The MPA NR sites do not show an increase in *S. purpuratus* density compared to the non-MPA NR and cannot support the hypothesis that it is acting as refugia in response to wasting.

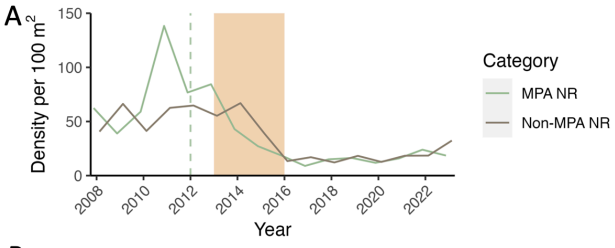


Figure 14. (A) Annual mean density per 100 m^2 of *M. franciscanus* from 2008–2023 at the non-MPA natural reef (Non-MPA NR) and Marine Protected Area natural reef (MPA NR) sites before and after the wasting event of 2013-2016. (B) Best-fit spatiotemporal model for *M. franciscanus* based on progressive-change BACIPS method. Green dashed lines indicate MPA imposition in 2012. Orange bars indicate the Blob warming period and wasting event from 2013-16.

In 2008 and 2009, the annual mean density of *M. francisanus* at non-MPA NR sites increased from 40.6/100 m^2 to 66.2/100 m^2 before returning to 41.2/100 m^2 in 2010. The annual mean density of *M. franciscanus* at non-MPA sites varied between 40.6/100 m^2 and 66.9/100 m^2 from 2008 to 2015 and declined to 13.4/100 m^2 in 2016. There was no increase until 2022-2023 when density increased to 32.3/100 m^2. For the MPA NR sites, annual mean density increased from s 62.3/100 m^2 to 138.3/100 m^2, from 2008 to 2011. The subsequent years exhibited a steady annual decline to 8.9/100 m^2 in 2017. Density does not increase nor decrease largely in the years post-disturbance, following a similar trend to non-MPA NR sites, yet increased minorly in 2021-2023 from 15.8/100 m^2 to 18.5/100 m^2 (Fig. 14A). Following the wasting event, the annual mean density of *M. franciscanus* gradually declined each year (Fig. 14B), resulting in the selection of the step-change model (ω_{step} = 62.1%, *P*-value = 0.097). As this was not statistically significant, the next best model was asymptotic ($\omega_{asymptotic}$ = 46.1%). Neither site category is showing an increase in *M. franciscanus* and future change is unlikely from likely

progressive-change models. MPA NR and non-MPA NR are not acting as refugia for *M. fraciscanus*.

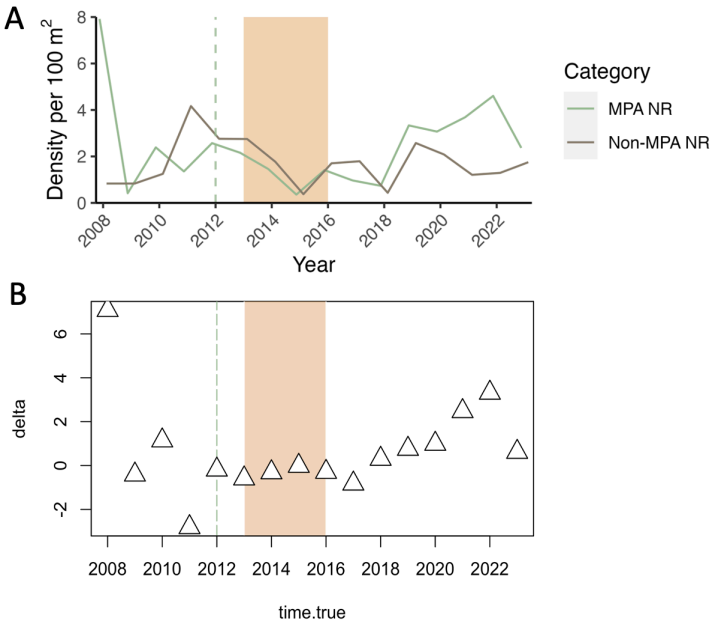


Figure 15. (A) Annual mean density per 100 m^2 of *A. parvimensis* from 2008–2023 at the non-MPA natural reef (Non-MPA NR) and Marine Protected Area natural reef (MPA NR) sites before and after the wasting event of 2013-2016. (B) Best-fit spatiotemporal model for *A. parvimensis* based on progressive-change BACIPS method. Green dashed lines indicate MPA imposition in 2012. Orange bars indicate the Blob warming period and wasting event from 2013-16.

Annual mean density for *A. parvimensis* at non-MPA NR sites increased from 0.8/100 m^2 to 4.2/100 m^2 from 2008-2011. A decline coinciding with the Blob occurred in 2015 when density declined to 0.4/100 m^2, with annual variation between 1.2-2.6/100 m^2 in the following years. At MPA NR sites, *A. parvimensis* mean density declined from 7.9/100 m^2 to 0.8/100 m^2 in 2008-2009. Through the warming disturbance, the mean density declined in 2015 to 0.3/100 m^2. Post-warming, the annual mean density at MPA NR surpasses that of the non-MPA NR and peaks at 4.6/100 m^2 in 2022 (Fig. 15A). Following the wasting event, the annual mean density of *A. parvimensis* slightly declined (Fig. 15B), resulting in the selection of the linear-change model (ω_{linear} = 52.7%, *P*-value = 0.486). The next best model is step-change (ω_{step} = 41.7%). There is an

increase in annual mean density for *A. parvimensis* at non-MPA NR sites, in addition to likely linear and step-change models. These species appear in higher densities at the MPA NR from 2019 onwards, indicating a preference for this habitat.

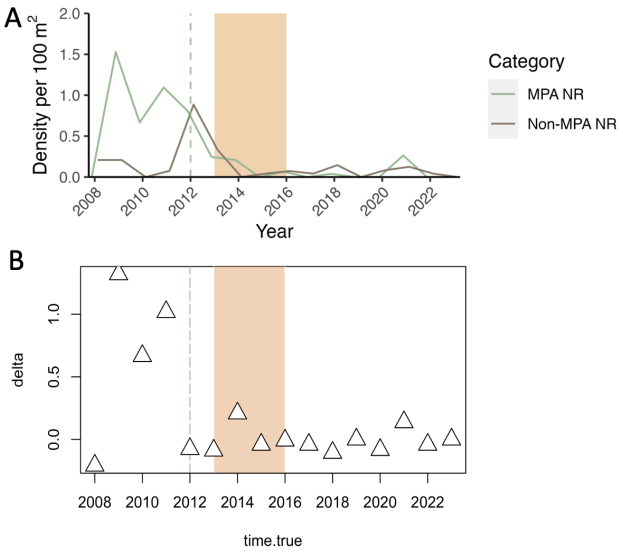


Figure 16. (A) Annual mean density per 100 m^2 of *A. californicus* from 2008–2023 at the non-MPA natural reef (Non-MPA NR) and Marine Protected Area natural reef (MPA NR) sites before and after the wasting event of 2013-2016. (B) Best-fit spatiotemporal model for *A. californicus* based on progressive-change BACIPS method. Green dashed lines indicate MPA imposition in 2012. Orange bars indicate the Blob warming period and wasting event from 2013-16.

The annual mean density for *A. californicus* at non-MPA NR sites varied between 0.0/100 m^2 and 0.9/100 m^2 from 2008-2012. From 2014 through 2023, annual mean density remained below 0.1/100 m^2. At MPA NR sites, *A. californicus* increased from 0.0/100 m^2 to 1.5/100 m^2 from 2008-2009. A gradual decline followed from 1.7-/100 m^2 to 0.0/100 m^2 (Fig. 16A). Following the wasting event, the annual mean density of *A. californicus* declined abruptly (Fig. 16B), resulting in the selection of the step-change model (ω_{step} = 85.8%, *P*-value = 0.015). *A. californicus* does not show signs of recovery at either site category, inferring support for the alternate hypothesis.

Discussion

Echinoderm populations have suffered significantly following the Blob warming event (2013-16), as annual mean density consistently declined over the years 2013-16 with minimal signs of recovery over surveyed sites. An increase in density was observed most clearly at PVR-adjacent sites and throughout the natural reef from 2019 onward. Sea stars were most susceptible to the Blob and did not recover. *P. giganteus* and *P. ochraceus* were most affected and have shown no signs of recovery post-environmental disturbance. Adversely, *P. miniata* did not completely decline, exhibiting signs of resilience, and was seen at similar densities at both PVR-adjacent and natural reef sites. The sea cucumbers, *A. parvimensis* and *A. californicus* were affected to a lesser extent but both are observed at higher mean density in 2023, 6 years post-disturbance. Drastic reductions of sea urchin densities were observed during the Blob, though *M. franciscanus* made a stronger recovery than *S. purpuratus* at both PVR-adj and NR though at a lower density[6].

MPA NR and non-MPA NR sites both suffered from drastic declines in annual mean density for all taxa besides *A. parvimensis,* showing signs of regular annual fluctuation and potentially a recovery trend at MPA NR sites from 2018 onwards. Despite this, MPA NR is not acting as a refuge habitat for these species following the Blob, nor did it observe an increase in density as habitats recover in the years post-disturbance. Both habitats were equally impacted by the Blob and the taxa did not exhibit a significant recovery. It is possible to consider that for more slowly reproducing taxa, low density per area is exhibited as the introduction of novel habitats may cause organisms to migrate to nearby habitats without an equal rate of population growth.

As species recovery was observed at PVR-adjacent sites, future directions in population recovery involve looking specifically at population density trends on the PVR artificial reef modules. Sea stars and urchins exhibited signs of observable adult migration to these modules as shown in the VRG's preliminary data from initial invertebrate density surveys from 2020 and 2021 from the months following reef implementation (S2). The population density of *P. ochraceus, P. giganteus,* and *M. franciscanus* increased in the years following the implementation of the PVR module in 2020, reaching their peak mean density per 100 m^2 in 2022 and 2023 (S2). *P. miniata* and *A. parvimensis* were observed annually from 2019-2023 and *A. californicus* has not been observed on the artificial reef (S2). Conservation efforts are better oriented to the attributes of the newly created artificial reef, compared to the protection of existing habitats, and may provide the levels of nutrients and refuge needed to support threatened populations in reproduction and long-term recovery. Habitat restoration has the most potential at sites with degraded reefs compared to construction far from existing structures[16]. Furthermore, though it is too early to observe ecology spillover, increased density amongst PVR modules shows promise for the recovery of neighbouring populations due to juvenile recruitment and migration.

Comparable outcomes in coral reef restoration efforts in the Gulf of Mannar in southeast India have been shown in response to global bleaching events of 2014-2017. A significant increase in coral diversity and density was observed on the modules one year post-implementation and aided long-distance dispersal of juveniles to nearby habitats as well as long-term stability of habitat and species habitation[19]. This type of long-term effect and long-range impact due to artificial reef influence is what we would like to see in temperate

habitats along the Southern California Bight and reefs near PVR seem to be progressing in that direction.

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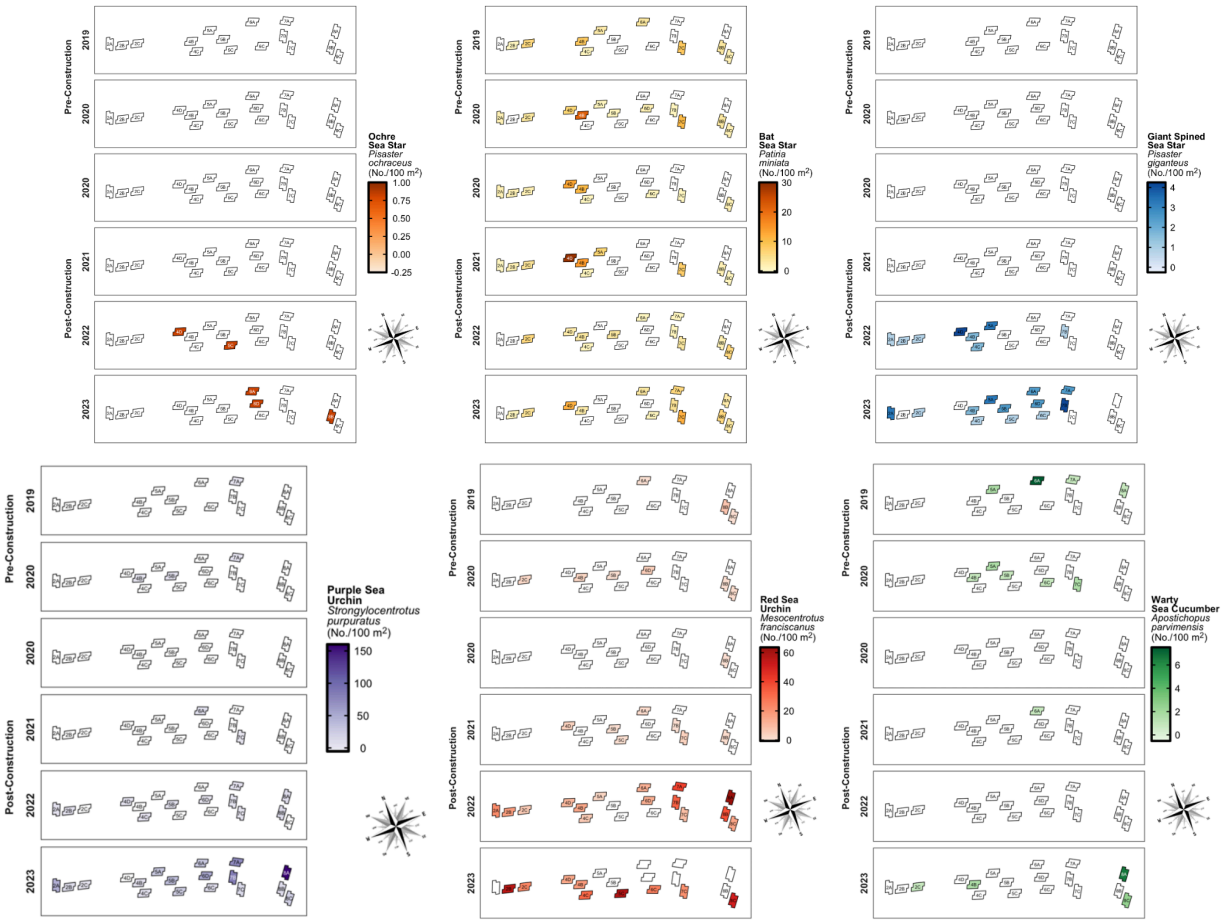
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Supplementary Figures

Supplementary Table 1. Comprehensive model output for all focal species and relative likelihood percentages for progressive-change BACIPS between PVR and MPA NR sites compared to the natural reef (inclusive of the non-MPA NR).

	PVR		MPA NR	
Species	**Model**	**Likelihood**	**Model**	**Likelihood**
P. ochraceus	Step-Change	80.5%	Step-Change	66.4%
P. miniata	Step-Change	61.7%	Step-Change	67.2%
P. giganteus	Step-Change	94.3%	Step-Change	77.9%
S. purpuratus	Step-Change	75.1%	Step-Change	52.4%
M. fransciscanus	Linear	56.6%	Step-Change	62.1%
A. parvimensis	Linear	95.7%	Linear	52.7%
A. californicus	Linear	63.3%	Step-Change	85.8%

Supplementary Figure 2. The annual mean density / 100m^2 of focal species at Palos Verdes Reef artificial reef modules from 2019 through 2023. PVR implementation was 2020. *A. californicus* was not observed.



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